J Autism Dev Disord (2014) 44:883–893
DOI 10.1007/s10803-013-1941-5

ORIGINAL PAPER

Use of a Peer Support Intervention for Promoting Academic Engagement of Students with Autism in General Education Settings

Erin E. McCurdy · Christine L. Cole

Published online: 22 October 2013
© Springer Science+Business Media New York 2013

Abstract Students with autism spectrum disorder (ASD) have been shown to benefit from being educated in general education classrooms that provide interactions with typically developing peers. However, behaviors exhibited by students with ASD frequently lead to their return to segregated special education settings. Evidence-based interventions that are both cost-efficient and easy to use in general education settings are needed. The purpose of this study was to evaluate the effects of a simple peer support intervention on the minor disruptive, off-task behaviors of three elementary students with high-functioning ASD in three different general education classrooms. Results indicated the peer support intervention was effective in reducing the off-task behaviors of the students with ASD in these inclusion settings. Practical implications and directions for future research are discussed.

Keywords Peer support · Inclusion · Autism spectrum disorder · General education

Introduction

The most recent revision of the Individuals with Disabilities Education Improvement Act (IDEA 2004) mandates that children with disabilities be educated in general education classrooms with their typical peers to the maximum extent possible. Additionally, the law states that children with disabilities may only be removed from regular education and placed in special education when, due to the nature or severity of the disability and with the appropriate supports and services, they cannot receive a satisfactory education in the general education classroom (IDEA 2004).

Inclusive education has become a particularly important issue in the education of children with autism spectrum disorders (ASD). ASD is a developmental disorder characterized by difficulties in communication and social behavior, and the presence of restricted and repetitive/challenging behaviors and interests (Mash and Wolfe 2002). Inclusion is viewed by many as an appropriate practice for educating children with ASD because of the potential benefits including the increased opportunity for social interactions with nondisabled peers, the possibility of developing friendships, the gains in communication, social, and adaptive behavior skills, and the participation in age-appropriate activities that may enhance social competence and ultimately lead to successful post-school adjustment (Hunt et al. 1986; Hunt and McDonnell 2007; McDonnell et al. 1998).

Despite the benefits of inclusion, there are also barriers that arise during the implementation of inclusive practices. One of the initial barriers is gaining the support of the school administration and staff. Many schools never consider placing students with ASD in general education because of the provision in IDEA 2004 that allows for the removal of children with disabilities from general education setting if the nature or severity of the child's disability inhibits his or her learning. Other schools may require the child to "earn" his or her way into general education, thereby refusing to allow the child access to the general classroom (Merrell et al. 2006).

E. E. McCurdy · C. L. Cole
School Psychology Program, Lehigh University, Bethlehem, PA 18018, USA

E. E. McCurdy (✉)
617 South Rose Street, Baltimore, MD 21224, USA
e-mail: eem206@lehigh.edu

Another potential barrier to the inclusion of children with ASD in general education classrooms is the presence of disruptive or off-task behaviors. Many students with ASD exhibit loud or repetitive behaviors, and have high levels of inattention. Disruptive, off-task behaviors exhibited by students with ASD often interrupt the instructional environment and result in decreased opportunities for learning, not only for the target student, but for the entire class. When these behaviors are demonstrated by a child with ASD, the result may be a decision by school personnel to place the student in special education (Arceneaux and Murdock 1997; Sterling-Turner et al. 2001).

A variety of interventions have been used to address the disruptive, off-task behaviors displayed by children with ASD such as positive reinforcement, visual schedules, social stories, and self-monitoring. *Positive reinforcement* in the form of prizes, verbal praise, tokens to purchase desired items, and food has been used as a means of increasing desired behaviors (i.e., academic engaged time), and decreasing problem behaviors (i.e., disruptive, off-task behavior) in children with ASD. Reinforcement is provided by an adult or a peer when the child displays appropriate behavior or refrains from displaying inappropriate behaviors (Kay et al. 2006; Mangus et al. 1986; Matson and Boisjoli 2009). *Visual schedules* and *Priming* have also been used to decrease problem behaviors (Banda and Grimmet 2008; Crosland and Dunlap 2012; Dettmer et al. 2000; Dooley et al. 2001; Lequia et al. 2012). The interventions are based on the rationale that if the child knows what to expect next, he or she may be less anxious and therefore not engage in disruptive behavior. Alternatively, the child may remain on task to get to the next, more desired, activity. Similar to positive reinforcement, there is empirical evidence to support that visual schedules and priming can successfully reduce problem behaviors in children with ASD (Crosland and Dunlap 2012; Dettmer et al. 2000; Dooley et al. 2001).

Self-monitoring has also been used as a means of behavior change with students with ASD (Koegel et al. 2012). With self-monitoring, the child is taught to observe and record instances of his or her own behavior, with the goal of reducing the disruptive actions. Self-monitoring has demonstrated positive effects in reducing disruptive behaviors in children with ASD and other disabilities (e.g., Koegel et al. 2012; Koegel et al. 1992; Newman et al. 1997).

Finally, another intervention strategy that has been used to reduce off-task and disruptive behaviors in students with ASD is *Prevent-Teach-Reinforce* (PTR; Iovannone et al. 2009). PTR is a team-based approach that includes conducting a functional behavior assessment (FBA) to determine the function of the behavior, and then creating an intervention plan based on the results of the FBA. Results of multiple studies have shown that PTR increases appropriate social skills and academic engagement as well as reduce problem behaviors (Crosland and Dunlap 2012; Iovannone et al. 2009; Strain et al. 2011).

Although each of these intervention strategies has produced reductions in disruptive, off-task behaviors in children with ASD, they are primarily adult-mediated interventions and/or provide little opportunity to engage with peers. Involving peers rather than adults in intervention efforts, termed peer-mediated interventions, may have several advantages over adult-directed intervention strategies. First, peers are readily available across a variety of settings. In any classroom, there are several peers who may be willing to assist the student or implement an intervention. Another advantage of involving peers is that they naturally affect each other's behavior (Fowler 1988; Hoff and Robinson 2002). Since peers are present in many settings throughout the school day they may also help to promote maintenance and generalization of positive behavior change. The peers themselves may serve as cues for the child with disabilities to remember the appropriate behaviors he or she learned during the intervention (Hoff and Robinson 2002). An additional benefit of using peers is that they can effectively meet the students' needs while saving valuable teacher time. Using a peer to assist in an intervention allows the teacher to spend more time teaching and programming for students' needs, which is both time and cost effective (Christensen et al. 2004; Hoff and Robinson 2002). Peer-mediated interventions have also been shown to increase active engaged time and performance feedback. If an intervention provides students with increased opportunities to respond and to receive feedback, it may result in a higher level of attention (Hoff and Robinson 2002). Finally, using peers in classroom interventions resulted in academic, cognitive, affective, and social gains across heterogeneous groups of peers and individuals receiving the intervention (Hoff and Robinson 2002).

Although peer-mediated interventions have routinely been used to target academic skills including individual and class-wide peer tutoring and cooperative learning groups (Carter et al. 2005; Dugan et al. 1995; Hunt et al. 1994; Kamps et al. 1994, 1995; Ward and Ayvazo 2006), very few peer-mediated interventions have targeted disruptive behaviors. One exception is a simple peer support strategy that has been shown to decrease disruptive, off-task behaviors displayed by children in the classroom is peer prompting of appropriate behaviors. Arceneaux and Murdock (1997) provided an example of this strategy in their investigation of the effects of prompting by a familiar peer on the loud, birdlike noises made by a boy with developmental disabilities during silent reading times. In this study, a typical peer who got along well with other

students and enjoyed helping peers was taught by the general education teacher to prompt the student with disabilities to focus on the academic activity whenever he made birdlike noises. During silent reading, the peer pulled her desk next to the student with disabilities and, when he emitted the noises, she pointed to the book he was reading to focus his attention.

An ABAB design was used to evaluate the effects of this intervention. During baseline, the student with disabilities emitted an average of 16.67 sounds per session. When the intervention was implemented, the number of sounds decreased to an average of 7 per session. A return to baseline showed a slight increase in sounds and, in the final phase, he only made an average of 2.60 sounds per session. Follow-up data at 3-1/2 and 5 weeks showed only 0.67 and 0.4 sounds per session, respectively.

Involving peers to promote desired behaviors of students with ASD in inclusive settings may be particularly beneficial, as the intervention encourages involvement with typical peers, addressing the core social skills deficits of students with ASD. Additionally, pairing with a peer removes the stigmatization of having a paraprofessional hover over the child in a general education setting. Finally, if the student's behavior can be successfully redirected by a peer, the student may be included in more classes, even when a paraprofessional is unavailable. Despite the possible benefits of involving peers and the positive outcomes of the Arceneaux and Murdock (1997) investigation, no studies were found that examined the use of peer prompting to reduce disruptive behaviors of individuals with ASD. Therefore, the purpose of this study was to extend the work of Arceneaux and Murdock (1997) by evaluating the effects of this simple peer support intervention on the disruptive, off-task behavior of students with ASD in general education settings.

Method

Participants and Setting

Three males with ASD who were enrolled in two elementary schools in an urban school district on the east coast participated in the study. Jack was an 8-year-old diagnosed with ASD who participated in a typical 3rd-grade classroom full time with minimal supports in School One. Jack had average intellectual abilities and used verbal communication. Jack required minimal accommodations to complete his work that included shorter assignments, breaks, and continuous reminders from adults of what he was supposed to be doing. According to his teacher, no peers in the classroom identified Jack as their friend, but Jack believed everyone was his friend and enjoyed interacting with peers. Jack's teacher referred him for intervention due to his high-rate off-task behaviors during center time including playing in his desk, crawling on the floor, talking with peers, walking aimlessly around the room, and drawing instead of paying attention. His teacher stated that in addition to his off-task behavior interfering with his work, it was disruptive to the other students. She stated that he often crawled under the feet of other students and would talk to peers during work time, distracting them from their work. During center time, Jack was expected to complete independent seatwork in math and writing while the teacher led small reading groups in the back of the room.

Chase was a 7-year-old boy with a diagnosis of ASD who participated in a typical 2nd-grade classroom full time with no additional supports in School Two. Similar to Jack, Chase had average intelligence and used verbal communication. Chase did not require any modifications to complete work other than near-constant adult reminders to keep working. Chase also had limited appropriate interactions with peers, and his teacher stated that peers were annoyed when he was in their work group because he often caused them to get in trouble due to high rates of off-task behavior. Chase was referred for intervention due to his high-rate off-task behaviors during center time. Chase's behaviors included laying on the floor, talking to peers, reading off-task material, and staring out the window. The teacher stated that the laying on the floor and talking to peers were the most disruptive behaviors. Chase would lie on the floor in areas where the other students needed to work and would yell at them if they disrupted him. Chase's expectations during center time were the same as Jack's.

Scott was an 11-year-old boy diagnosed with ASD who participated in a typical 5th-grade classroom with minimal supports in School Two. Scott had average intelligence and used verbal communication. He received pull out support services for a learning disability in both math and reading. Similar to Jack and Chase, Scott had few appropriate interactions with peers and the teacher stated he had no friends in his classroom. Unlike the other participants, Scott did not readily seek out interactions with peers. Scott was referred for intervention due to his high rate off-task behaviors during Guided Reading. Scott's behaviors included playing in his desk and with materials, laying his head down and closing his eyes, and engaging in self-stimulatory behaviors (e.g., flicking/tapping his pencil and fingers in the air and rubbing the wire-binding of notebooks). The teacher stated that the pencil tapping and rubbing the wire binding of the notebooks were very distracting when she was teaching because of the noises it made. She stated that she had to continuously prompt Scott to stop engaging in this behavior that was disruptive to her teaching. During guided reading, Scott was expected to

follow along with a story read by the teacher from the overhead projector to the entire class and then complete any independent seatwork assigned after the story was read.

Once the target students were identified, each teacher was asked to nominate three typical students as possible peer supporters. Students were considered eligible if the teacher believed they did not need reminders to stay on task, and their learning or work completion would not be adversely affected by participating in the intervention. If the peer gave their assent to participate, they were then asked to take home a permission form for their parent's signature. The first peer from each class to return the signed permission form was trained as the peer supporter.

Jack's peer supporter was Daniel. He was a 9-year-old boy whose teacher reported performed well academically, was a leader in the class, and enjoyed helping other students when they needed it. Chase's peer supporter was Allison. She was an 8-year-old girl who was considered by the teacher to be a top student in the class, but was very quiet and tended to lead by example. Scott's peer supporter was Tiana. She was an 11-year-old girl nominated by her teacher who reported she was on grade level, academically.

All sessions were conducted in the students' general education classrooms during the academic periods their teachers had identified as having the highest levels of off-task behaviors. There were approximately 28 students and 1 teacher in each classroom, with no instructional aide.

Dependent Measures, Observational Procedures, and Interobserver Agreement

Off-Task Behavior

The primary dependent measure was *off-task behavior*, individually defined for each student. For Jack and Chase, off-task behavior was defined as doing any work other than the assigned work, playing with materials in or on the desk, talking to peers, talking to the peer supporter about something other than the assigned work, staring out the window, crawling or laying on the floor, and walking around the classroom. For Scott, off-task behavior was defined as looking away from the overhead screen for more than 5 s, putting his head down on the desk, playing with materials, talking to peers, and reading material other than the assigned reading.

Off-task behavior was assessed through direct observation of each participant in the general education classroom 5 days per week. The data collector entered 5 min after the class period began and recorded data for a total of 20 min each day. Data were recorded for off-task behavior with the Behavioral Observation of School Students (B.O.S.S; Shapiro, 2003) using a 15-s partial-interval procedure. For comparison purposes, data were also recorded for the off-task behavior of typical peers every 5th interval using the same 15-s partial-interval procedure (Shapiro 2003). To gain an average, overall view of peer performance, a different peer was observed during each peer interval. For example, during the first peer observation interval, the student in the front right corner of the room was observed; then during the second peer observation interval, the student directly behind the first peer was observed. Data collection rotated throughout the room in this manner for the entire observation session. Peer off-task behavior included talking to other peers about topics unrelated to the assignment, out-of-seat behavior, and passively not attending to assigned work.

Direct observations were conducted by trained graduate students who were enrolled in doctoral programs in Special Education or School Psychology, and were naïve to the purpose of the study. Observers sat in the back of the room, where they had a clear view of the student and their assigned area or desk. Prior to the study, data collectors received training in the use of the 15-s interval recording system. Training consisted of practice using the recording system via videotapes of students engaged in academic tasks in the classroom until each achieved 80 % agreement with the investigator on the dependent measure.

Interobserver Agreement (IOA)

For the purpose of determining IOA, a second observer simultaneously, but independently, observed 25 % of the sessions across all phases of the study. IOA was calculated for off-task behavior by determining the number of agreements, divided by the number of agreements plus disagreements and multiplying the result by 100. Mean IOA for off-task behavior across all sessions was 93.3 (range = 76.9–100 %). No booster sessions were required as IOA did not fall below 75 %.

Peer Supporter Questionnaire

A second dependent variable was the *Peer Opinion Questionnaire*), an examiner-constructed questionnaire that assessed typical peers' perspectives of the student with ASD (i.e., do you like the target student, is the target student your friend, would you play with the target student). The investigator conducted a private interview with each peer supporter twice during the study. The first interview was conducted during initial peer training; the second interview, with four additional questions (i.e., did this experience change your opinion of the target student), was held immediately after the study ended.

Table 1 Teachers' mean responses to the *intervention rating profile-15*

Question	Mean response
1. This was an acceptable intervention to decrease off-task behavior	5.6
2. Most teachers would find this intervention appropriate for decreasing off-task behavior	5.0
3. This intervention was effective in reducing the student's off-task behavior	5.3
4. I would recommend the use of this intervention to other teachers	5.0
5. The student's off-task behavior was severe enough to warrant use of this intervention	6.0
6. I would be willing to use this intervention with other students	5.6
7. This intervention does not result in negative side-effects for students	6.0
8. This intervention would be appropriate for a variety of students	6.0
9. This intervention is consistent with those I have used before in the classroom	4.3
10. The intervention is a fair way to handle the student's off-task behaviors	6.0
11. This intervention is reasonable for off-task behavior	6.0
12. I like the procedures used in this intervention	5.3
13. This intervention was a good way to handle the student's off-task behavior	5.6
14. Overall, this intervention was beneficial for the student	6.0

Intervention Acceptability

A modified version of the Intervention Rating Profile—15 (IRP-15; Martens et al. 1985) was used to assess teachers' perceived acceptability of the intervention. At the conclusion of the study, all three teachers completed the form which asks teachers to indicate their degree of agreement or disagreement with each statement using a 6-point likert scale (see Table 1). The measure has been found to be reliable with a Cronbach's alpha of .98 (Martens et al. 1985).

To evaluate the target students' and peer supporters' perceived acceptability of the intervention, each student was given a modified version of the Children's Intervention Rating Profile (CIRP; Turco and Elliott 1986) following the final intervention period. The six-item CIRP also uses a 6-point likert scale to indicate degree of agreement or disagreement with each of the statements (see Table 2). The CIRP has been used in previous research and has been found to be reliable with a coefficient alpha of .89.

Experimental Design and Data Analysis

A multiple-baseline design across three participants was used to evaluate the effects of the peer support intervention on the disruptive, off-task behaviors of the three students with ASD. In this study, the intervention was introduced for the first participant when an increasing trend in off-task behavior was evident. Following the implementation of the peer support intervention with the first participant, a decreasing trend was established prior to introducing the intervention to the next participant. This was repeated for the third participant.

Data were analyzed using visual inspection of graphically displayed data to examine means, trends, and level changes. In addition, percentage of non-overlapping data (PND; Scruggs and Mastropieri 1994) was also calculated for each participant. PND was calculated by counting the number of points in the intervention condition that fell outside the range of values for the baseline condition, and dividing that number by the number of data points in the intervention condition and multiplying by 100. Higher PND is associated with a greater impact on the target behavior.

Procedures

Baseline

During baseline, typical classroom procedures remained in effect. The teacher was asked to seat the student with ASD

Table 2 Target students' and peer supporters' mean responses to the *children's intervention rating profile*

Question	Mean response target student	Mean response peer supporters
1. My supporter was fair./My buddy was fair	1.0	1.0
2. My supporter caused problems for me in the classroom./My buddy caused problems for me in the classroom	6.0	6.0
3. There are better ways to handle my distractions than the buddy system./There are better ways to handle my buddy's distractions than the buddy system	5.6	6.0
4. The buddy system would be good to use with other kids./The buddy system would be good to use with other kids	1.0	1.3
5. I liked having a buddy to help me to remember to do my work./I liked helping my buddy remember to do his work	1.0	1.6
6. I think that having a buddy would help other kids to remember to do their work	1.0	1.0

and the peer supporter in close proximity to each other. No other changes were made to typical classroom conditions.

Assessment of Opinions

Following baseline and prior to training, the peer supporter was interviewed individually by the investigator in a quiet location using the *Peer Opinion Questionnaire.*

Peer Training

Immediately following the interview, individual training of the peer supporter was conducted during free time within the school day in a quiet room with the peer, the investigator, and an assistant. The assistant was an instructional aide employed by the school who was familiar with the student with ASD and the targeted behaviors. The investigator first defined for the peer in simple terms what it means to be off-task during class. The assistant then role played examples of off-task behavior and other desired behaviors, and the investigator helped the peer identify the off-task behaviors. The peer was then asked to identify the off-task behaviors without the help of the investigator.

Once the peer was able to identify the assistant's off-task behavior with 100 % accuracy on 10 consecutive trials, he or she was taught how to prompt desired behavior. To begin, the investigator demonstrated the appropriate way to prompt while the assistant displayed the off-task behaviors. Each time the assistant displayed off-task behaviors, the researcher lightly tapped the materials or the desk. The investigator also modeled for the peer how to give nonverbal positive feedback such as thumbs up and smiles for encouragement. The peer was told to give nonverbal feedback whenever he felt the target student was doing a good job. After watching the demonstrations, the peer took the place of the investigator, prompting the assistant and providing positive feedback when the assistant was not engaging in off-task behavior.

Once the peer demonstrated appropriate prompting with 100 % accuracy on 10 consecutive trials, the peer was trained to provide verbal feedback at the end of the session. In the same way, the primary investigator first modeled how to provide feedback to the assistant and next the peer practiced giving feedback to the assistant. When the peer successfully gave feedback with no help from the investigator on 5 consecutive trials, training was considered completed. All peer supporters were trained using the same training procedures and in one training session. Each training session was conducted during a nonacademic period and lasted no more than 30 m (detailed training procedures are available by request from the first author).

Peer Support Intervention

Following peer training, the student with ASD and the peer supporter continued to sit next to each other during the targeted general education class period. At the beginning of each class period, the peer supporter reminded the student with ASD of the desired behaviors and gave him verbal encouragement (e.g., "I know you can do it."). Then, whenever the student with ASD displayed the target off-task behavior, the peer prompted the student as taught during training. The teacher was asked to informally monitor the peer supporter throughout the class period. If the redirection was unsuccessful after five attempts by the peer, the teacher was asked to redirect the student with ASD, and then move away from the dyad. Teacher involvement was not required for any of the participants during the study. When the student remained on-task, the peer provided nonverbal forms of encouragement (e.g., a thumbs up, a smile, a head nod). At the end of each intervention session, the peer and the student with ASD met in a quiet corner of the room or outside in the hall so the peer could deliver feedback and encouragement to the student with ASD. For example, following the period, the peer prompter may have said to the student with ASD, "Good job remembering to bring your math book the work table, but tomorrow we need to pay attention to our math page instead of playing with the eraser. I know you can do it!"

Peer Support Plus MotivAider

For Tiana, the peer who was supporting Scott, an additional component was added to the intervention following initial implementation of the peer support intervention. During implementation of the intervention, the investigator observed that Tiana failed to prompt Scott when he engaged in off-task behavior. The first time this was observed, she was retrained using procedures similar to those provided during the initial peer training. However, after the investigator observed this same behavior a second time, Tiana reported to the primary investigator that she was forgetting to look at Scott. The primary investigator then gave the peer supporter a MotivAider set to vibrate every 2 min to prompt her to observe Scott's behavior. No further training sessions were required.

Post Assessment

Following completion of the study, the peer supporters' were interviewed using the Peer Supporter's Post-intervention Interview. The researcher met with each student during a nonacademic period and asked each of the questions. Additionally, social validity was assessed using the

IRP-15 and CIRP. The teachers completed the IRP-15 on their own time and returned it to the researcher. The teachers also sat with the target students and the peer supporters individually and helped them to complete the CIRP by reading the questions to each student.

Results

Off-Task Behavior

Data for the off-task behavior of each participant with ASD are shown in Fig. 1. For Jack, the mean for off-task behavior during baseline was 68.74 %, but levels decreased to a mean of 13.02 % during the peer support intervention condition. Jack's off-task behavior during intervention was slightly lower than that of his typical classmates (M = 18.75 %). The peer support intervention produced an immediate change in level that maintained for the duration of the intervention. The PND for Jack was 100 %.

For Chase, off-task behavior decreased from a baseline mean of 73.18 % to a mean of 15.00 % during the peer support intervention condition. Similar to Jack, Chase's level of off-task behavior also fell below that of his classroom peers (M = 19.17 %). An immediate change in level was observed from the final baseline session to the

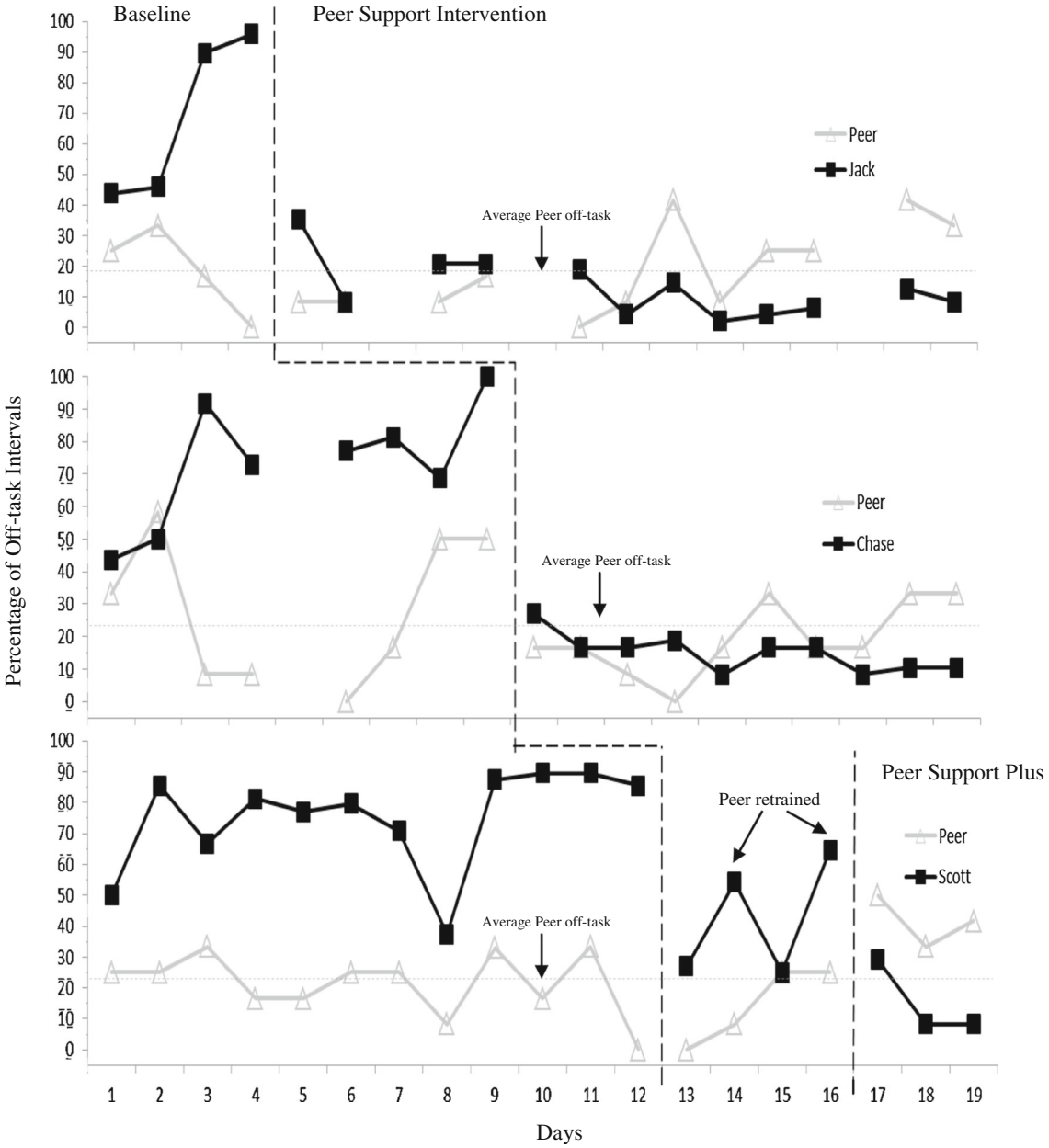


Fig. 1 The percentage of intervals of off-task behavior for the target students and peer supporters

first peer support intervention session, and the decrease in off-task behavior continued for the remainder of the intervention condition. The PND for Chase was 100 %.

For Scott, off-task behavior decreased from a baseline mean of 75.39 % to a mean of 42.71 % during the peer support intervention condition. Scott's off-task behavior was higher than that of his peers' (M = 14.58) during this phase and the behavior demonstrated a high degree of variability. When the MotivAider was added, another decrease was observed, to a mean of 15.28 %, with a continuing downward trend. During this condition, Michael's off-task behavior was lower than that of his typical classmates (M = 41.67). While the PND for the peer support condition was only 50 %, this improved to 100 % during the peer support plus MotivAider condition.

Peer Supporter Questionnaire

Results of the *Peer Opinion Questionnaire* were similar from baseline to post-treatment for all peer supporters. At both the baseline and post-treatment assessments, all three peers reported they liked the student with ASD, he was a friend, and they would play with him. All peer supporters stated that the experience did not alter their opinion of the target student, they liked the experience, and they would do it again. Despite no change in peer opinions, there was a change in the spontaneous teacher report of the target students. All teachers reported increased comfort in giving the student with ASD independent work to complete because, as off-task behavior decreased, each student's work completion increased. These statements were made spontaneously to the investigator at the completion of the study

Treatment Acceptability

Results of the IRP-15 and the CIRP showed that all teachers, target students, and peer supporters found the intervention to be acceptable and appropriate for decreasing off-task behavior (see Table 2). Overall, the results across individuals and questions were positive.

Discussion

The peer support intervention was effective in reducing the off-task behavior of students with ASD to a level similar to that of their classroom peers. Although it was necessary to add a prompt to remind Scott's peer supporter to attend to his off-task behavior, no other changes were made to the intervention. The results of the current study replicate and extend the work of Arceneaux and Murdock (1997) by demonstrating that this simple peer support intervention can be effective in reducing off-task behaviors of students with ASD in general education settings who are as young as 7 years old. The lack of change in the peer comparison data across all the participants suggests that change in the target students' behavior was most likely due to the peer support intervention.

In addition to quickly reducing off-task behavior, the intervention was highly acceptable to teachers, the target students, and the peers who supported them. Jack and Chase's teachers both independently commented they believed the intervention also would be effective with other students in their classroom who were not diagnosed with ASD, but who displayed high rates of off-task behavior. Additionally, Jack and Chase both made comments to the effect that they really enjoyed working with the peers, and that they considered the peers to be "new friends."

The setting for Jack and Chase was a period in which the majority of the class was instructed to work independently or in pairs while the teacher led a small reading group. For both boys, it was especially difficult because they had almost no teacher supervision, and if they did become slightly off-task, there was no one to bring them back on-task with a quick reminder. Typically then, these students spent the majority of the period engaging in off-task behavior and completing almost no work. With the peer support intervention, the peer supporter was able to redirect the students' attention to the task before they became too distracted from their work. By the end of the study, both peer supporters reported they no longer needed to prompt Jack or Chase. They reported that, if both boys started to become distracted, as soon as they looked at their peer supporter, they returned immediately to their work.

For Scott, the setting for the intervention was a Guided Reading class in which the teacher turned off the lights, read to the students from an overhead projector, and provided little opportunity for class participation. It was easy for Scott to become distracted as his reading level was not at the same level of the material being read to the class so, if Scott fell behind while following along, he was unable to find his place on his own. Also, the activity during this period may not have been a good match for Scott and others. In fact, during the intervention, Scott's peer supporter reported that at times it was difficult for her to pay attention during Guided Reading, even though she was identified as the student who exhibited the highest levels of on-task behavior. When the primary investigator inquired as to why she had difficulty paying attention, the peer supporter reported that it was boring just listening to her teacher read. Thus, the activity itself may have contributed to the initial poor results, as the peer supporter was not paying attention and prompting the student with ASD consistently.

Another issue that arose with Scott and not with Jack or Chase was a concern expressed by his peer supporter that the other students were watching her. When the peer was being trained to use the MotivAider, the primary investigator asked the peer if she had any concerns and she reported she was nervous that the other students were looking at her. The primary investigator assured her this was not the case and, if anyone had any questions, she could tell them that she was helping the teacher with a special project. The peer supporter's self-consciousness may have been due to the fact that she and Scott were in 5th grade, while the other two participants were in 2nd and 3rd grade. A 5th-grader may be more likely to be self-aware and self-conscious using this intervention, suggesting it may be more appropriate for younger students.

Limitations and Future Directions

There are several possible limitations of the current study. The first and most obvious is that only a small number of students with ASD participated. The students were chosen because they were engaging in high rates of disruptive, off-task behavior and were included in general education settings with minimal supports. All three students were able to participate in the general education curriculum with minimal accommodations and had no intellectual disabilities. In addition to being higher functioning, two of the three participants were socially motivated and sought out peers in the classroom for social interactions, which may have increased the effectiveness of the intervention because of the increased proximity to peers. Additionally, the target off-task behaviors of the focus students were relatively minor in severity and never escalated to a severe level of disruption. These characteristics of the participants and their behaviors suggest that the effectiveness of the intervention may be limited to high functioning individuals with minor problem behaviors. More research is needed to determine if the intervention would produce similar positive outcomes with other students on the spectrum.

In addition to participant characteristics, there are limitations in the type of data collected during the study. As this was an initial evaluation of the intervention, data collection was solely focused on the effects of the intervention on disruptive, off-task behavior. No data were collected on work completion of the student with ASD or the typical peer, nor was intervention integrity assessed. These are important areas that should be evaluated in future research to determine whether the intervention is being delivered by peers appropriately and has a meaningful effect on the academic success of the focus students and their peers.

Another concern was the small number of sessions during Scott's final intervention condition. The limited number of sessions was due to time constraints imposed by standardized testing and the end of the school year. Although the data showed a downward trend and the PND was 100 %, it is possible that, with more sessions, the rate of behavior may have increased. Additional research involving longer intervention phases and post-intervention follow-up may help to answer the question regarding the durability of the positive outcomes.

In addition, no FBA was conducted to determine the function of the students' off-task behaviors. In this study, the intervention was applied with no knowledge of why each student was engaging in off-task behaviors. It is possible that off-task behaviors with different functions may respond differently to the intervention. Future research should include an FBA to determine if there is a relationship between function and responsiveness to the peer support intervention.

A final limitation of the current study is that the intervention was evaluated solely as a multi-component package. There was no additional evaluation to determine whether the intervention as a whole was effective or whether certain parts of the intervention were responsible for the change in the disruptive, off-task behavior of the student with ASD. In future research, the parts of the intervention could be examined separately to piece out the contributions of each component (i.e., does just having the meeting at the beginning and end of the period reduce off-task behavior, does just prompting alone reduce off-task behavior).

Future research should also examine whether the peer support intervention is effective with younger students. The intervention was very successful with both the 2nd- and 3rd-grade students, and the self-consciousness that arose in the 5th-grade peer supporter was not an issue for the younger peer supporters. It is possible that the peer support intervention may be most effective for younger (e.g., kindergarten or 1st-grade) students, and future research should explore this.

Another direction that should be examined is the relationship that may develop as a result of the pairing of the student with ASD and the peer supporter. Although there was no change in peer opinions, no direct measure of behavior was recorded in regard to their friendship. Research should examine whether different pairings of students with ASD and peer supporters results in more interactions in other settings, such as lunch or recess.

Finally, future research should also examine if the effects of the peer support intervention generalize to other class periods. The student with ASD's off-task behavior should be observed to determine if they learn to stay on-task without prompting. The peer supporter's behavior also should be observed to determine if the peer continues to prompt the student with ASD at other times.

Overall, the results of the study showed this simple peer support intervention was effective in quickly reducing the disruptive off-task behaviors of three students with ASD in their inclusive general education classrooms. The intervention was simple and easy to use, and was viewed as acceptable by the teachers, the students with ASD, and the peer supporters. Minimal training was required to implement the intervention and, therefore, most teachers could probably implement the intervention on their own. However, due to the comments made by the 5th-grade peer supporter, the intervention may be best utilized with younger students.

References

Arceneaux, M. C., & Murdock, J. Y. (1997). Peer prompting reduces disruptive vocalizations of a student with developmental disabilities in a general eighth-grade classroom. *Focus on Autism and Developmental Disabilities, 12*, 182–186.

Banda, D. R., & Grimmet, E. (2008). Enhancing social and transition behaviors of persons with autism through activity schedules: A review. *Education and Training in Developmental Disabilities, 43*, 324–333.

Carter, E. W., Cushing, L. S., Clark, N. M., & Kennedy, C. H. (2005). Effects of peer support interventions on students' access to the general curriculum and social interactions. *Research and Practice for Persons with Severe Disabilities, 30*, 15–25.

Christensen, L., Young, K. R., & Marchant, M. (2004). The effects of a peer-mediated positive behavior support program on socially appropriate classroom behavior. *Education and Treatment of Children, 27*, 199–234.

Crosland, K., & Dunlap, G. (2012). Effective strategies for the inclusion of children with autism in general education classrooms. *Behavior Modification, 36*, 251–265.

Dettmer, S., Simpson, R. L., Smith-Myles, B., & Ganz, J. B. (2000). The use of visual supports to facilitate transitions of students with autism. *Focus on Autism and Other Developmental Disabilities, 15*, 163–169.

Dooley, P., Wilczenski, F. L., & Torem, C. (2001). Using an activity schedule to smooth school transitions. *Journal of Positive Behavior Interventions, 3*, 57–61.

Dugan, E., Kamps, D., Leonard, B., Watkins, N., Rheinberger, A., & Stackhaus, J. (1995). Effects of cooperative learning groups during social studies for students with autism and fourth-grade peers. *Journal of Applied Behavior Analysis, 28*, 175–188.

Fowler, S. A. (1988). The effects of peer-mediated interventions on establishing, maintaining and generalizing children's behavior changes. In R. H. Horner, G. Dunlap, & R. L. Koegel (Eds.), *Generalization and maintenance* (pp. 143–170). Baltimore, MD: Paul H. Brooks Publishing Co.

Hoff, K. E., & Robinson, S. L. (2002). Best practices in peer-mediated interventions. In A. Thomas & J. Grimes (Eds.), *Best practices in school psychology IV* (pp. 1555–1567). Kent, Ohio: National Association of School Psychologists.

Hunt, P., Goetz, L., & Anderson, J. (1986). Quality of IEP objectives associated with placement on integrated versus segregated school sites. *Journal of the Association for Persons with Severe Handicaps, 11*, 125–130.

Hunt, P., & McDonnell, J. (2007). Inclusive education. In S. Odom, R. H. Horner, M. E. Snell, & J. Blacher (Eds.), *Handbook of developmental disabilities* (pp. 269–291). New York, NY: Guilford Press.

Hunt, P., Staub, D., Alwell, M., & Goetz, L. (1994). Achievement by all students within the context of cooperative learning groups. *Journal of the Association for Persons with Severe Handicaps, 19*, 290–301.

Individuals with Disabilities Education Act Amendments of 2004, 11 Stat. 37 U.S.C. Section 1401 (2004).

Iovannone, R., Greenbaum, P., Wang, W., Kincaid, D., Dunlap, G., & Strain, P. (2009). Randomized control trial of a tertiary behavior intervention for students with problem behaviors: Preliminary outcomes. *Journal of Emotional and Behavioral Disorders, 17*, 213–225.

Kamps, D. M., Barbetta, P. M., Leonard, B. R., & Delquadri, J. (1994). Classwide peer tutoring: An integration strategy to improve reading skills and promote peer interactions among students with autism and general education peers. *Journal of Applied Behavior Analysis, 27*, 49–61.

Kamps, D. M., Leonard, B., Potucek, J., & Garrison-Harrell, L. (1995). Cooperative learning groups in reading: An integration strategy for students with autism and general classroom peers. *Behavioral Disorders, 21*, 89–109.

Kay, S., Harchik, A. E., & Luiselli, J. K. (2006). Elimination of drooling by an adolescent student with autism attending public high school. *Journal of Positive Behavior Interventions, 8*, 24–28.

Koegel, L. K., Koegel, R. L., Hurley, C., & Frea, W. D. (1992). Improving social skills and disruptive behavior in children with autism through self management. *Journal of Applied Behavior Analysis, 25*, 341–353.

Koegel, L., Matos-Freden, R., Lang, R., & Koegel, R. (2012). Interventions for children with autism spectrum disorders in inclusive school settings. *Cognitive and Behavioral Practice, 19*, 401–412.

Lequia, J., Machalicek, W., & Rispoli, M. J. (2012). Effects of activity schedules on challenging behavior exhibited in children with autism spectrum disorders: A systematic review. *Research in Autism Spectrum Disorders, 6*, 480–492.

Mangus, B., Henderson, H., & French, R. (1986). Implementation of a token economy by peer tutors to increase on-task physical activity time of autistic children. *Perceptual and Motor Skills, 63*, 97–98.

Martens, B. K., Witt, J. C., Elliot, S. N., & Darveaux, D. X. (1985). Teacher judgments concerning the acceptability of school-based interventions. *Professional Psychology: Research and Practice, 16*, 191–198.

Mash, E. J., & Wolfe, D. A. (2002). *Autism and childhood-onset schizophrenia. Abnormal child psychology* (pp. 257–291). Belmont, CA: Wadsworth.

Matson, J. L., & Boisjoli, J. A. (2009). The token economy for children with intellectual disabilities and/or autism: A review. *Research in Developmental Disabilities, 30*, 240–248.

McDonnell, J., Thorson, N., & McQuivey, C. (1998). The instructional characteristics of inclusive classes for elementary students with severe disabilities: An exploratory study. *Journal of Behavioral Education, 8*, 415–437.

Merrell, K. W., Ervin, R. A., & Gimpel, G. A. (2006). *School psychology for the 21st century: foundations and practices*. New York: Guilford Press.

Newman, B., Tuntigian, L., Ryan, C. S., & Reinecke, D. R. (1997). Self-management of a DRO procedure by three students with autism. *Behavioral Interventions, 12*, 149–156.

Scruggs, T. E., & Mastropieri, M. A. (1994). The utility of the PND statistic: A reply to Allison and Gorman. *Behaviour Research and Therapy, 32*, 879–883.

Shapiro, E. S. (2003). *Behavioral observation of students in schools-BOSS (Computer Software)*. San Antonio, TX: Psychological Corporation.

Sterling-Turner, H. E., Robinson, S. L., & Wilczynski, S. M. (2001). Functional assessment of distracting and disruptive behaviors in the school setting. *School Psychology Review, 30*, 211–226.

Strain, P. S., Wilson, K., & Dunlap, G. (2011). Prevent-Teach-Reinforce: Addressing problem behaviors of students with autism in general education classroom. *Behavior Disorders, 36*, 160–171.

Turco, T. L., & Elliott, S. N. (1986). Students' acceptability ratings of interventions for classroom misbehaviors: A study of well-behaving and misbehaving youth. *Journal of Psychoeducational Assessment, 4*, 281–289.

Ward, P., & Ayvazo, S. (2006). Classwide peer tutoring in physical education: Assessing its effects with kindergartners with autism. *Adapted Physical Activity Quarterly, 23*, 233–244.